

Volume Vi Linear Algebra Reference

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Linear Algebra

0.1 Vector Space Preliminaries

Definition 0.1.1 (List). Suppose n is a non-negative integer. A **list** of length n is an ordered collection of n elements.

Definition 0.1.2 ($\mathbf{F}^n$). Let $\mathbf{F}$ denote $\mathbb{R}$ or $\mathbb{C}$, and let n be a positive integer. Then $\mathbf{F}^n$ is defined to be the set of all lists of length n of elements of $\mathbf{F}$:

$$\mathbf{F}^n = \{(x_1, \dots, x_n) : x_1, \dots, x_n \in \mathbf{F}\}.$$

The entries in a list in $\mathbf{F}^n$ are called the coordinates of the list.

Definition 0.1.3 (Addition in $\mathbf{F}^n$). Let

$$x = (x_1, \dots, x_n), \quad y = (y_1, \dots, y_n)$$

be vectors in $\mathbf{F}^n$. The sum of x and y is defined by

$$x + y = (x_1 + y_1, \dots, x_n + y_n).$$

Theorem 0.1.4 (Addition Is Commutative in $\mathbf{F}^n$). *For all $x, y \in \mathbf{F}^n$,*

$$x + y = y + x.$$

Go to proof.

Definition 0.1.5 (Zero Vector in $\mathbf{F}^n$). The symbol 0 denotes the list of length n all of whose coordinates are 0 :

$$0 = (0, \dots, 0) \in \mathbf{F}^n.$$

Definition 0.1.6 (Additive Inverse in $\mathbf{F}^n$). Let

$$x = (x_1, \dots, x_n) \in \mathbf{F}^n.$$

The additive inverse of x is defined by

$$-x = (-x_1, \dots, -x_n).$$

Definition 0.1.7 (Scalar Multiplication in $\mathbf{F}^n$). Let

$$\lambda \in \mathbf{F} \quad \text{and} \quad x = (x_1, \dots, x_n) \in \mathbf{F}^n.$$

The product of λ and x is defined by

$$\lambda x = (\lambda x_1, \dots, \lambda x_n).$$

Definition 0.1.8 (Vector Space). A **vector space** over $\mathbf{F}$ is a set V together with an addition on V and a scalar multiplication

$$\mathbf{F} \times V \rightarrow V$$

such that the following properties hold for all $u, v, w \in V$ and all $a, b \in \mathbf{F}$:

- (i) $u + v \in V$;
- (ii) $u + v = v + u$;
- (iii) $(u + v) + w = u + (v + w)$;
- (iv) there exists $0 \in V$ such that $v + 0 = v$;
- (v) for every $v \in V$ there exists $-v \in V$ such that $v + (-v) = 0$;
- (vi) $av \in V$;
- (vii) $a(u + v) = au + av$;
- (viii) $(a + b)v = av + bv$;
- (ix) $a(bv) = (ab)v$;
- (x) $1v = v$.

Definition 0.1.9 (Vector and Point). An element of a vector space V is called a **vector**. When $V = \mathbf{F}^n$, an element of $\mathbf{F}^n$ may also be called a **point**.

Definition 0.1.10 (Real Vector Space). A **real vector space** is a vector space over $\mathbb{R}$.

Definition 0.1.11 (Complex Vector Space). A **complex vector space** is a vector space over $\mathbb{C}$.

Theorem 0.1.12 (Unique Additive Identity). *A vector space has a unique additive identity. Go to proof.*

Theorem 0.1.13 (Unique Additive Inverse). *Every vector in a vector space has a unique additive inverse. Go to proof.*

Theorem 0.1.14 ($0v = 0$). *For every $v \in V$,*

$$0v = 0.$$

Go to proof.

Theorem 0.1.15 ($a0 = 0$). *For every $a \in \mathbf{F}$,*

$$a0 = 0.$$

Go to proof.

Theorem 0.1.16 ($(-1)v = -v$). *For every $v \in V$,*

$$(-1)v = -v.$$

Go to proof.

Definition 0.1.17 (Subspace). Let V be a vector space over $\mathbf{F}$. A subset $U \subseteq V$ is called a **subspace** of V if U is itself a vector space over $\mathbf{F}$ with the same addition and scalar multiplication as on V .

Theorem 0.1.18 (Conditions for a Subspace). *Let V be a vector space over $\mathbf{F}$ and let $U \subseteq V$. Then U is a subspace of V if and only if the following three conditions hold:*

- (i) $0 \in U$;
- (ii) for all $u, v \in U$, one has $u + v \in U$;
- (iii) for all $a \in \mathbf{F}$ and all $u \in U$, one has $au \in U$.

Go to proof.

Definition 0.1.19 (Sum of Subspaces). Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V . The **sum** of $U_1, \dots, U_m$ is the set

$$U_1 + \dots + U_m = \{u_1 + \dots + u_m : u_1 \in U_1, \dots, u_m \in U_m\}.$$

Theorem 0.1.20 (Sum of Subspaces Is the Smallest Containing Subspace). *Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V . Then $U_1 + \dots + U_m$ is a subspace of V containing each of $U_1, \dots, U_m$. Moreover, if W is a subspace of V containing each of $U_1, \dots, U_m$, then*

$$U_1 + \dots + U_m \subseteq W.$$

Go to proof.

Definition 0.1.21 (Direct Sum). Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that

$$V = U_1 + \dots + U_m.$$

Then V is called the **direct sum** of $U_1, \dots, U_m$ if each vector $v \in V$ can be written in the form

$$v = u_1 + \dots + u_m, \quad u_1 \in U_1, \dots, u_m \in U_m,$$

in exactly one way. In this case, we write

$$V = U_1 \oplus \dots \oplus U_m.$$

Theorem 0.1.22 (Condition for a Direct Sum). *Let V be a vector space over $\mathbf{F}$, and let $U_1, \dots, U_m$ be subspaces of V such that*

$$V = U_1 + \dots + U_m.$$

Then the following are equivalent:

(i) $V = U_1 \oplus \dots \oplus U_m$;

(ii) if $u_1 \in U_1, \dots, u_m \in U_m$ and

$$u_1 + \dots + u_m = 0,$$

then

$$u_1 = \dots = u_m = 0.$$

Go to proof.

Theorem 0.1.23 (Direct Sum of Two Subspaces). *Let U and W be subspaces of a vector space V . Then*

$$U + W = U \oplus W$$

if and only if

$$U \cap W = \{0\}.$$

Go to proof.

Proofs

Capstone